AI, ML, DL, and NLP – Simple Notes

A beginner-friendly text-only guide covering the major concepts, techniques, algorithms, and applications of Artificial Intelligence, Machine Learning, Deep Learning, and Natural Language Processing.

Purpose: These notes are designed for students preparing for university courses, interviews, lecturer demonstrations, and basic AI/ML/NLP projects.

1. Artificial Intelligence (AI)

Artificial Intelligence is the field of computer science concerned with building machines and software that can perform tasks that normally require human intelligence.

What can AI do?

- Understand information and make decisions.
- Learn from data or experience.
- Solve problems.
- Understand language.
- Recognize images, speech, and patterns.
- Plan actions and achieve goals.
- Interact with humans.

Types of AI

- **Narrow AI:** AI designed for a specific task, such as recommendation systems, voice assistants, or spam detection.
- **General AI:** A theoretical form of AI that would perform a wide range of intellectual tasks like a human.
- **Super AI:** A hypothetical AI that would exceed human intelligence in many areas.

Important AI Concepts

- Knowledge representation: storing information in a form that a computer can use.
- Reasoning: using available information to reach conclusions.
- Search: exploring possible solutions to a problem.
- Planning: deciding a sequence of actions to achieve a goal.
- Expert systems: systems that use rules and domain knowledge to solve specialized problems.
- Agents: systems that perceive their environment and take actions.
- Computer vision: AI for understanding images and videos.
- Natural Language Processing: AI for understanding and generating human language.
- Robotics: combining AI with machines that can sense and act in the physical world.

AI Agents

An AI agent receives information from its environment, processes that information, and performs an action. A simple agent can be described as: Perception → Decision → Action.

- Simple reflex agent: acts using current input.
- Model-based agent: maintains some information about the environment.
- Goal-based agent: chooses actions to reach a goal.
- Utility-based agent: chooses actions that maximize usefulness or benefit.
- Learning agent: improves its behavior using experience.

2. Machine Learning (ML)

Machine Learning is a subset of AI. Instead of explicitly programming every rule, we provide data and an algorithm that learns patterns from the data.

Basic idea: Data → Learning Algorithm → Model → Prediction or Decision.

Main Types of Machine Learning

Supervised Learning

In supervised learning, the training data contains input examples and their correct outputs, called labels.

- **Classification:** predicts a category, such as spam or not spam.
- **Regression:** predicts a numerical value, such as house price.

Unsupervised Learning

In unsupervised learning, the data does not contain target labels. The algorithm tries to discover patterns or structures.

- Clustering: grouping similar data points.
- Dimensionality reduction: reducing the number of features while preserving useful information.
- Association learning: finding relationships between items.

Semi-Supervised Learning

Semi-supervised learning uses a small amount of labeled data together with a larger amount of unlabeled data.

Reinforcement Learning

Reinforcement learning trains an agent by giving rewards for good actions and penalties for undesirable actions. The agent learns a policy for choosing actions.

- Agent: the learner.
- Environment: the world in which the agent operates.
- State: the current situation.
- Action: what the agent does.
- Reward: feedback received after an action.
- Policy: strategy used to select actions.

Common Machine Learning Algorithms

- Linear Regression: predicts a continuous numerical value.
- Logistic Regression: commonly used for classification.
- Decision Tree: makes decisions using a tree of conditions.
- Random Forest: combines many decision trees.
- Support Vector Machine: finds a boundary that separates classes.
- K-Nearest Neighbors: predicts using nearby training examples.
- Naive Bayes: probabilistic classifier based on Bayes' theorem and a conditional independence assumption.
- K-Means: groups data into K clusters.
- Gradient Boosting: builds models sequentially to reduce prediction errors.
- XGBoost and LightGBM: efficient gradient boosting methods widely used for structured data.

Machine Learning Workflow

- 1. Define the problem.
- 2. Collect data.
- 3. Clean the data.
- 4. Explore the data.
- 5. Select or create features.
- 6. Split data into training, validation, and test sets.
- 7. Train the model.
- 8. Tune hyperparameters.
- 9. Evaluate the model.
- 10. Deploy and monitor the model.

Features and Labels

A feature is an input variable used by a model. A label or target is the output that the model is expected to predict.

Training, Validation, and Testing

- Training set: used to learn model parameters.
- Validation set: used to tune the model and select settings.
- Test set: used for final evaluation on unseen data.

Overfitting and Underfitting

- **Overfitting:** the model learns the training data too closely and performs poorly on new data.
- **Underfitting:** the model is too simple to learn the important patterns.
- A good model should generalize well to unseen data.

Bias and Variance

Bias is error caused by overly simple assumptions. Variance is error caused by a model being too sensitive to the training data. Good machine learning tries to balance bias and variance.

Common Evaluation Metrics

- Accuracy: proportion of correct predictions.
- Precision: among predicted positive cases, how many are actually positive.
- Recall: among actual positive cases, how many were correctly identified.
- F1-score: harmonic mean of precision and recall.
- Mean Absolute Error (MAE): average absolute prediction error.
- Mean Squared Error (MSE): average squared prediction error.
- Root Mean Squared Error (RMSE): square root of MSE.
- R^2: measures how well a regression model explains variation in the target.

3. Deep Learning (DL)

Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers to learn complex patterns from large amounts of data.

Basic idea: Input $\rightarrow$ Multiple Neural Network Layers $\rightarrow$ Output.

Artificial Neural Network

An artificial neural network is inspired by the basic idea of biological neurons. It contains interconnected computational units called neurons.

- Input layer: receives the input features.
- Hidden layers: learn intermediate patterns.
- Output layer: produces the final prediction.

Neuron

A neuron receives inputs, multiplies them by weights, adds a bias, and passes the result through an activation function.

A simple mathematical form is: z = w1x1 + w2x2 + ... + b.

Activation Functions

- Sigmoid: produces values between 0 and 1.
- Tanh: produces values between -1 and 1.
- ReLU: outputs zero for negative values and the input for positive values.
- Leaky ReLU: allows a small negative output instead of setting all negative values to zero.
- Softmax: converts multiple output scores into a probability distribution.

Forward Propagation

During forward propagation, input data moves from the input layer through the hidden layers to the output layer. The network calculates a prediction.

Loss Function

A loss function measures how different the model prediction is from the correct answer. Training tries to minimize this loss.

- Mean Squared Error is common for regression.
- Binary Cross-Entropy is common for binary classification.
- Categorical Cross-Entropy is common for multi-class classification.

Backpropagation

Backpropagation calculates how much each model parameter contributed to the error. The calculated gradients are then used to update the weights.

Gradient Descent

Gradient descent is an optimization method that updates model parameters in the direction that reduces the loss.

Basic idea: calculate loss $\rightarrow$ calculate gradients $\rightarrow$ update weights $\rightarrow$ repeat.

Common Deep Learning Architectures

- MLP: basic feed-forward neural network.
- CNN: especially useful for images and spatial patterns.
- RNN: designed for sequential data.
- LSTM: an RNN architecture designed to handle longer-term dependencies.
- GRU: a simpler gated recurrent architecture.
- Transformer: uses attention mechanisms and is highly effective for language and many other sequence tasks.
- Autoencoder: learns to reconstruct input data and can be used for representation learning or anomaly detection.
- GAN: uses a generator and discriminator in a competitive learning setup.

Deep Learning Training Terms

- Epoch: one complete pass through the training dataset.
- Batch size: number of examples processed before an update.
- Learning rate: controls the size of parameter updates.
- Optimizer: method used to update model parameters.
- Adam: a widely used adaptive optimization algorithm.
- Dropout: randomly disables some neurons during training to reduce overfitting.
- Batch normalization: normalizes intermediate activations and can improve training stability.
- Early stopping: stops training when validation performance stops improving.

4. Natural Language Processing (NLP)

Natural Language Processing is a field of AI that enables computers to process, understand, analyze, and generate human language.

NLP Applications

- Text classification.
- Sentiment analysis.
- Spam detection.
- Fake news detection.
- Machine translation.
- Question answering.
- Text summarization.
- Information extraction.
- Named Entity Recognition.
- Chatbots and virtual assistants.
- Speech and language applications.
- Search and information retrieval.

Basic NLP Pipeline

- 1. Collect text.
- 2. Clean the text.
- 3. Normalize the text.
- 4. Tokenize the text.
- 5. Convert text into numerical representations.
- 6. Train or apply a model.
- 7. Evaluate the results.

Text Preprocessing

- **Lowercasing:** converting text to a consistent case when appropriate.
- **Tokenization:** splitting text into tokens such as words, subwords, or characters.
- **Stop-word removal:** removing common words when they are not useful for the task.
- **Stemming:** reducing words to a rough root form.
- **Lemmatization:** reducing a word to its meaningful base form.
- **Cleaning:** removing unwanted HTML, URLs, punctuation, or other noise when appropriate.
- **Normalization:** standardizing text into a consistent representation.

Text Representation

- Bag of Words: represents text using word occurrence counts.

- TF-IDF: gives higher importance to words that are frequent in a document but less common across the collection.
- Word embeddings: represent words as dense numerical vectors.
- Contextual embeddings: produce representations that depend on surrounding words.
- Sentence embeddings: represent an entire sentence or passage as a vector.

Important NLP Models

- Naive Bayes and Logistic Regression are common traditional text-classification methods.
- RNNs process sequences step by step.
- LSTMs and GRUs handle sequential dependencies using gates.
- Transformers use attention to model relationships between tokens.
- BERT is a transformer-based model designed mainly for understanding language.
- GPT-style models are transformer-based generative language models.
- T5 treats many NLP tasks as text-to-text problems.

Attention Mechanism

Attention allows a model to focus on the most relevant parts of an input when producing an output. It is a central idea behind modern transformer models.

Transformer

A transformer uses attention mechanisms instead of relying primarily on sequential recurrence. Important concepts include self-attention, multi-head attention, positional information, encoder layers, and decoder layers.

- Encoder: commonly used to build representations of input text.
- Decoder: commonly used to generate output sequences.
- Encoder-decoder models can transform one sequence into another.
- Decoder-only models are widely used for text generation.

BERT and Fine-Tuning

BERT is commonly adapted to a specific task by fine-tuning it on task-specific labeled data. For example, BERT can be fine-tuned for sentiment classification or fake news classification.

Named Entity Recognition

NER identifies entities in text, such as people, organizations, locations, dates, and other domain-specific entities.

Sentiment Analysis

Sentiment analysis determines the emotional or opinion-related orientation of text, such as positive, negative, or neutral.

Text Classification

Text classification assigns a predefined category to a piece of text. Examples include spam detection, topic classification, toxicity detection, and fake news classification.

5. Relationship Between AI, ML, DL, and NLP

AI is the broad field concerned with intelligent computer systems. Machine Learning is one approach for building AI systems by learning from data. Deep Learning is a type of Machine Learning based on multi-layer neural networks. NLP is a field concerned with human language and can use traditional ML, deep learning, and modern transformer models.

In simple terms: AI is the broad area; ML is a major way to achieve AI; DL is a powerful ML approach; NLP is a language-focused field that can use ML and DL.

6. Generative AI

Generative AI creates new content such as text, images, audio, video, or code based on learned patterns.

- Large Language Models are generative models trained on large amounts of text.
- Prompt: the instruction or input given to a generative model.
- Token: a unit of text processed by a language model.
- Inference: using a trained model to generate an output.
- Fine-tuning: adapting a pretrained model to a particular task or domain.
- RAG: Retrieval-Augmented Generation combines retrieval of relevant information with language generation.

7. Retrieval-Augmented Generation (RAG)

RAG is a technique in which a system retrieves relevant information from a knowledge source and provides that information to a language model before generating an answer.

- Document loading: obtain documents from files or other sources.
- Chunking: divide documents into manageable pieces.
- Embedding: convert text into numerical vectors.
- Vector database: store and search vector representations.
- Retriever: find relevant chunks for a query.
- Prompt: combine the user question with retrieved context.
- Generator: an LLM generates the final answer.

RAG is useful when an application needs answers based on specific documents or frequently changing information.

8. Important AI/ML/DL/NLP Terms

- **Dataset:** collection of examples used for analysis or model development.
- **Model:** a learned mathematical representation used to make predictions or generate outputs.

- **Parameter:** a value learned during model training.
- **Hyperparameter:** a setting chosen before or during training, such as learning rate or tree depth.
- **Feature engineering:** creating or transforming input variables to improve learning.
- **Inference:** using a trained model to produce predictions.
- **Generalization:** the ability of a model to perform well on unseen data.
- **Embedding:** a numerical vector representation of an object such as a word, sentence, image, or document.
- **Transfer learning:** using knowledge learned from one task or dataset for another task.
- **Fine-tuning:** further training a pretrained model on a specific task or domain.

9. Common Interview Questions

What is AI?

AI is the field of creating computer systems that can perform tasks requiring abilities such as reasoning, learning, perception, language understanding, and decision-making.

What is Machine Learning?

Machine Learning is a subset of AI in which algorithms learn patterns from data and use those patterns to make predictions or decisions.

What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers to learn complex representations from data.

What is NLP?

NLP is the field of AI that enables computers to process, understand, analyze, and generate human language.

What is the difference between classification and regression?

Classification predicts categories or classes, while regression predicts numerical values.

What is overfitting?

Overfitting occurs when a model learns the training data too closely and does not generalize well to unseen data.

Why is data preprocessing important?

Preprocessing improves data quality and creates a suitable representation for learning. It can include cleaning, normalization, tokenization, feature preparation, and handling missing values.

What is an epoch?

An epoch is one complete pass through the training dataset.

What is an embedding?

An embedding is a numerical vector representation that captures useful information about an item such as a word, sentence, image, or document.

What is a transformer?

A transformer is a neural network architecture based heavily on attention mechanisms and is widely used for language understanding and generation.

10. Simple Learning Roadmap

- Step 1: Learn Python basics.
- Step 2: Learn mathematics for AI: linear algebra, probability, statistics, and basic calculus.
- Step 3: Learn AI fundamentals: search, agents, knowledge representation, and reasoning.
- Step 4: Learn Machine Learning and common algorithms.
- Step 5: Learn data preprocessing and model evaluation.
- Step 6: Learn neural networks and Deep Learning.
- Step 7: Learn NLP preprocessing and text representation.
- Step 8: Learn transformers, BERT, and modern language models.
- Step 9: Learn embeddings, vector databases, and RAG.
- Step 10: Build practical projects and evaluate them properly.

11. Final Summary

Artificial Intelligence is the broad concept of creating intelligent computer systems. Machine Learning allows systems to learn patterns from data. Deep Learning uses multi-layer neural networks to learn complex patterns. Natural Language Processing focuses on human language. Modern NLP increasingly uses transformer-based deep learning models. Together, these technologies form the foundation of many current applications such as chatbots, recommendation systems, computer vision, search engines, language translation, document question answering, and intelligent agents.

End of Notes